



> Intelligent navigation



> myExam Companion



> Elite performance



> Patient-friendly design

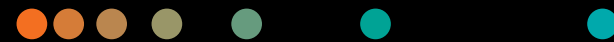


> Technology overview

SOMATOM X.ceed with myExam Companion

**Intelligent imaging.
Exceeding excellence.**

➤ siemens-healthineers.us/somatom-xceed



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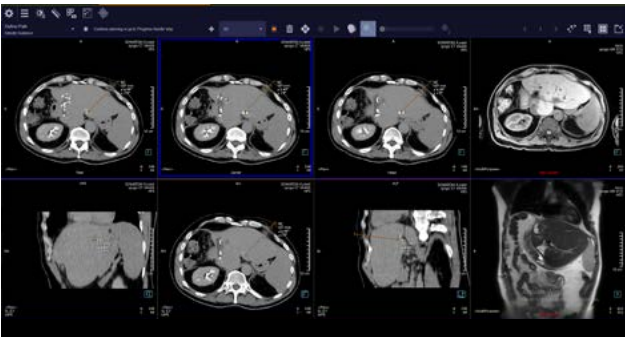
Intelligent navigation for enhanced consistency

As the number and complexity of radiological procedures increases, demands on staff are reaching unsustainable levels—impacting image quality. Too often, the full potential of CT scanners remains untapped.

SOMATOM® X.ceed is a high-resolution, high-speed, low-dose CT scanner with the category-best imaging chain.¹ With the unique guidance of myExam Companion and myNeedle Companion, it enables excellent clinical decisions in the most challenging clinical situations such as emergencies and advanced interventions.



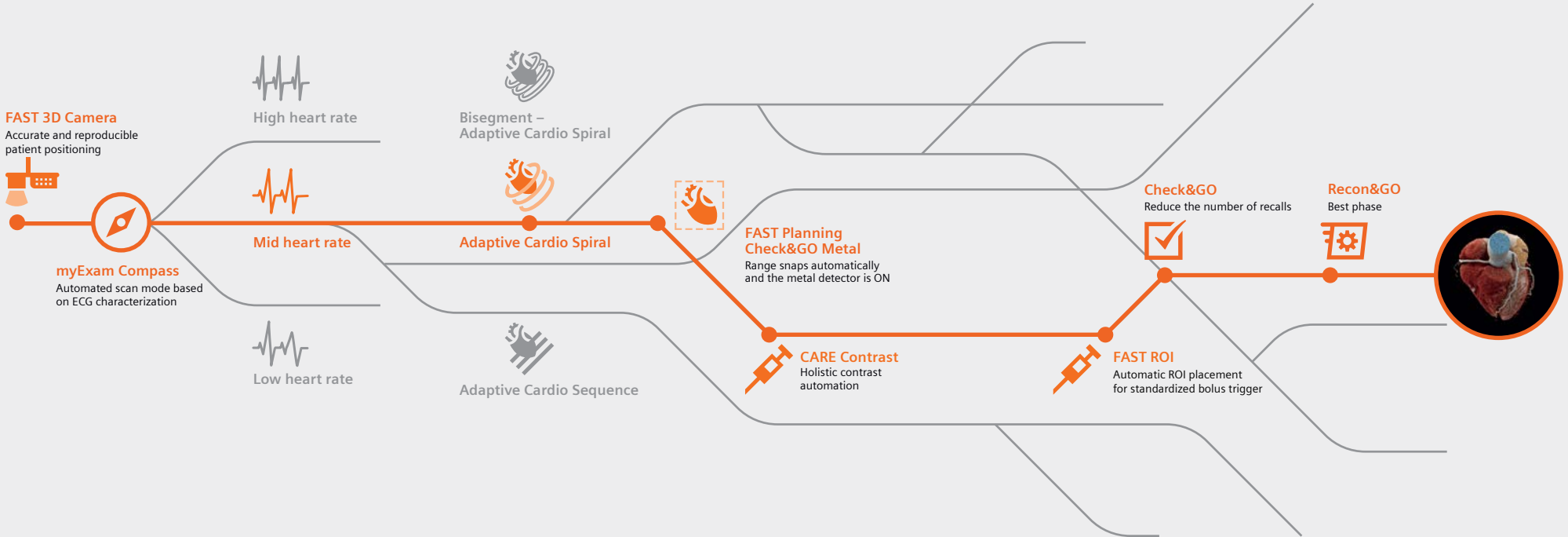
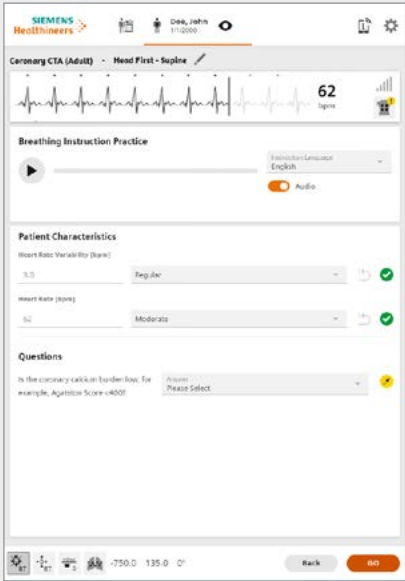
GO technologies powered by myExam Companion
Check&GO identifies potential errors in organ coverage, contrast media volume and distribution, and the presence of wearable metal objects (e.g., belts, necklaces) – helping users take immediate action or make corrections. By automating many postprocessing tasks, Recon&GO reduces the number of workflow steps: the results are automatically sent to PACS, including labeled CPR for the main vessels, ribs, and spine.



myNeedle Companion: Universal. Simple. Powerful.
myNeedle Companion harmonizes needle procedures across modalities. It helps simplify even complex image-guided needle procedures with targeted needle path planning and guidance, image fusion, and laser-guided needle insertion. Dedicated in-room control options offer radiologists the flexibility to work as they prefer—with or without an assistant.

myExam Companion: Intelligence that works with you

Reliable and reproducible results with myExam Companion
myExam Companion is a unique approach to scanner operation, designed to make work easier for users, personalize procedures for patients, and deliver consistent and comprehensive results for radiologists. It guides users of all experience levels through routine examinations as well as more complex procedures like stroke, spectral studies, or coronary CTA – for example, by asking the right question at the right time such as “Is the coronary calcium burden low?”, “Does the patient have stents?”, or “Are automatic reconstructions of the coronaries needed?” The answers are linked to predefined scan parameters.





Elite performance from door to diagnosis

SOMATOM X.ceed benchmarks the category with an optimal combination of high temporal and spatial resolution, power, and scan speed designed to master the most challenging clinical situations at low dose. myExam Companion offers radiology teams automated guidance throughout.

High native temporal resolution

The 0.25 s rotation time sets a new benchmark for single-source CT. A fast sampling rate of 8 kHz ensures that the scanner can handle the additional data.



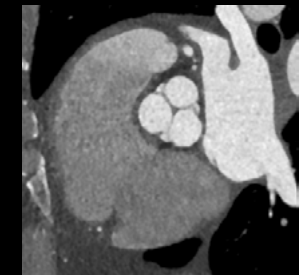
100 kV, CTDI_{vol} 30 mGy



70 kV, CTDI_{vol} 7.7 mGy
Cardiac CT with simultaneous high temporal and spatial resolution



70 kV, CTDI_{vol} 3.2 mGy
Enhanced differentiation of plaque types and detectability of contrast media in the most distal vessels



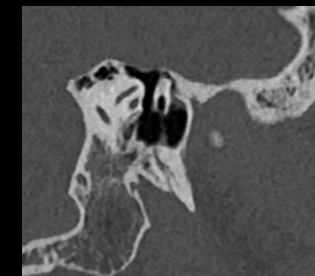
70 kV, CTDI_{vol} 4 mGy
High-quality raw data to reduce motion artifacts in coronaries, valves, and other heart structures

High spatial resolution

Even at full speed and low dose, thanks to the stability of the tiny focal spot (down to 0.4 x 0.5 IEC) and high detector density of 920 channels per row.



Sn130 kV, CTDI_{vol} 24.2 mGy



Sn130 kV, CTDI_{vol} 24.2 mGy
High-resolution imaging of the inner ear with lower dose thanks to Tin Filter



90 kV, CTDI_{vol} 4.3 mGy
High spatial resolution across the whole field of view even with fast rotation, and matching reconstruction of up to 1024 x 1024



High scan speed

261 mm/s thanks to fast detector sampling of 8 kHz that boosts image quality at the outer field of view, from head to toe even in large patients.



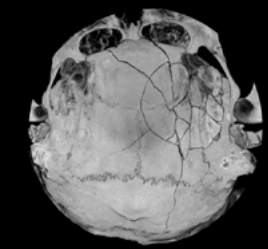
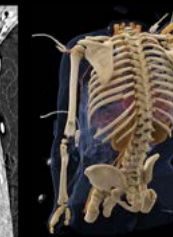
140 kV, CTDI_{vol} 3.9 mGy



90 kV, CTDI_{vol} 3.4 mGy
Excellent emergency imaging with simultaneous high temporal and spatial resolution



80 kV, CTDI_{vol} 2.36 mGy
Reduced motion artifacts with uncooperative patients such as children or emergencies



110 kV, CTDI_{vol} 60.6 mGy
Fast assessment of emergency imaging with zero-click trauma visualization*

High Vectron® tube power

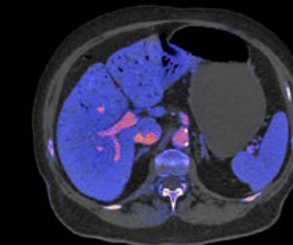
Reserves of up to 1,300 mA/120 kW enable dose-neutral Dual Energy and low-dose techniques even for large patients.



120 kV, CTDI_{vol} 12.1 mGy



70 kV, CTDI_{vol} 1.9 mGy
Reducing dose and contrast media while improving low-contrast resolution with low kV imaging



AuSn120 kV, CTDI_{vol} 9.42 mGy
Dose-neutral spectral imaging for a broad range of patients with 50 cm field of view



Sn100 – 150 kV, CTDI_{vol} 14.2 mGy

Patient-friendly design with an 82 cm bore

SOMATOM X.ceed is designed to transform how patients interact with both the user and the CT scanner—and how they perceive their care.

The large 82 cm bore is ...

- Ideal for obese patients and trauma, orthopedic, or interventional procedures
- Ideal to help both patients and users relax

The Mobile Workflow enables ...

- Staying close to the patient at all times by:
 - Preparing all scans right at the gantry tablets
 - Starting the scan remotely with the remote control
 - Returning to the patient right after the scan
 - Previewing images directly on the tablet
- Placement and charging at the docking station on the gantry or at your workplace

The FAST 3D Camera helps set the ...

- Right dose modulation with FAST Isocenter
- Correct body region with FAST Range
- Right scan direction with FAST Direction

The gantry-integrated Patient Observation Camera helps ...

- Monitor the patient even inside the gantry
- Keep an eye on the patient and spot even small movements

The gantry-integrated Visual Patient Instruction offers ...

- Improved compliance with breathing commands to potentially reduce motion artifacts
- Intuitive color-coded breath-hold countdown, displayed on the front and back of the tunnel
- Visual guidance, especially helpful for the hearing-impaired or patients who don't understand the local language

The gantry-mounted injector arm allows ...

- The injector to be positioned where you need it, when you need it
- The injector to be flexibly moved into other positions
- Work in a neat and patient-centered environment without a blocking injector cart

Technical specifications

Detector:	Stellar ^{Infinity} detector
X-ray tube:	Vectron® X-ray tube
Number of acquired slices:	128 slices
Rotation time:	up to 0.25 s
Native temporal resolution:	up to 125 ms
Generator power:	up to 120 kW
kV settings:	70 – 150 kV @ 10 kV Steps
Spatial resolution:	UHR mode 0.4 mm reconstructions
Max. scan speed:	261 mm/s
Table load:	up to 307 kg/676 lbs
Gantry opening:	82 cm



Technology overview

Vectron® X-ray tube

Delivers excellent low-kV properties and a small focal spot for high resolution:

- Up to 1,300 mA @ 70, 80, 90 kV
- Down to 0.4 x 0.5 (IEC) focal spot
- 70 – 150 kV in steps of 10 kV

Stellar^{Infinity} detector

Reduces image noise in every scan. Increased channel density and new geometry result in homogenous image quality, even in complex areas:

- TrueSignal technology with full electronic integration
- ADMIRE iterative reconstruction enables the generation of 0.5 and 0.4 mm slices
- Faster sampling of 8 kHz for improved spatial resolution with fast rotations (0.25 s)
- Precision Matrix enable high-resolution reconstructions of up to 768 x 768 and 1024 x 1024

Tin Filter

Cuts out lower energies to reduce dose and enhance image quality:

- Low-dose early detection in lung, colon, or calcium score exams
- Dose level comparable to that of a conventional X-ray system
- Tin-filtered topogram

TwinBeam Dual Energy

Acquires low- and high-kV datasets in a single scan:

- Rich diagnostic information that a conventional single source CT scan can't deliver
- Users can characterize, highlight, and quantify different materials for greater diagnostic confidence with virtually all patients

TwinSpiral Dual Energy

Provides both morphological and functional information in non-contrast exams:

- Datasets at two different energies
- Improved spectral separation with no dose penalty thanks to tin filtration of the high-kV scan
- New workflow concept of two scans integrated into a single acquisition

Flex 4D Spiral

Provides optimized dynamic acquisitions with continuously repeated bidirectional table movement during spiral acquisition:

- Extended range for 4D information
- Perfusion range of up to 11 cm in head applications, up to 22.5 cm in body applications, and dynamic CTA studies up to a scan range of 57.5 cm



At Siemens Healthineers, our purpose is to enable healthcare providers to increase value by empowering them on their journey toward expanding precision medicine, transforming care delivery, and improving patient experience, all enabled by digitalizing healthcare.

An estimated 5 million patients globally benefit every day from our innovative technologies and services in the areas of diagnostic and therapeutic imaging, laboratory diagnostics, and molecular medicine, as well as digital health and enterprise services.

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References

**syngo*.CT Extended Functionalities including Trauma Layouts is a medical device in its own right, WIP and not commercially available in the United States.

1. Data on file.

Courtesy of clinical images from left to right, page 4-5:

High native temporal resolution cases
Luzerner Kantonsspital, Luzern, Switzerland
Kantonsspital Baden, Baden, Switzerland
Diagnostikum Linz, Linz, Austria
Diagnostikum Linz, Linz, Austria

High spatial resolution cases
University Hospital Erlangen, Erlangen, Germany
University Hospital Erlangen, Erlangen, Germany
Kantonsspital Baden, Baden, Switzerland
Kantonsspital Baden, Baden, Switzerland

High scan speed cases
Luzerner Kantonsspital, Luzern, Switzerland
Kantonsspital Baden, Baden, Switzerland
Luzerner Kantonsspital, Luzern, Switzerland
University Hospital Erlangen, Erlangen, Germany
University Hospital Erlangen, Erlangen, Germany

High Vectron® tube power cases
University Hospital Erlangen, Erlangen, Germany
University Hospital Erlangen, Erlangen, Germany
University Hospital Erlangen, Erlangen, Germany
Kantonsspital Baden, Baden, Switzerland

Some of the features mentioned herein are optional.

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